ECB-RESTRICTED

EUROPEAN CENTRAL BANK

EUROSYSTEM

sSRyB calibration

Indicator-based approach

30/03/2023

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Motivation

- Construct and RRE-index that could be used together with other approaches to guide future activation and calibration of the sSyRB for RRE related risks
- Early warning models (EWM) can be used to detect vulnerable state pre-RRE-related systemic banking crises:
  - Identify top-performing indicators
  - Aggregate chosen indicators into an index for RRE risk (like SRI for cyclical risk)
- Purposes:
  - Provide information complimentary to SRI regarding the build-up of RRE vulnerabilities
  - Together with other approaches, could inform policy recommendations in the MPR and the ECB’s tasks under Article 5 SSMR
  - Basis for external communication following ECB decisions

=> The purpose of this presentation is to present the data, approach, provide early work in progress results and collect feedback

Literature

- Early warning models – rich literature using discrete dependent variable models, among others:
  - Barrell et al. (2010) used logit models to signal systemic banking crises
  - Demirgüç-Kunt and Detragiache (2005) used both the signalling approach and discrete dependent variable models to predict banking crises and to analyse their determinants
  - Lo Duca and Peltonen (2013), using discrete dependent variable models aimed at signalling systemic financial crises, showed that vulnerabilities build up non-linearly
  - Bruns and Poghosyan (2016) used a logit model aggregation, called extreme bound analysis, to signal fiscal stress events
- Most relevant references:
  - Ferrari, Cornacchia, Pirovano (2015) provide a formal statistical evaluation of potential early warning indicators for real-estate related banking crises – and highlight the important role of both real estate price variables and credit developments
  - Lang, Izzo, Fahr and Ruzicka (2019) propose a domestic cyclical systemic risk indicator capturing risks from domestic credit, real estate markets, asset prices, and external imbalances

Model used for the identification of early warning

We consider the following discrete choice (logit) model:

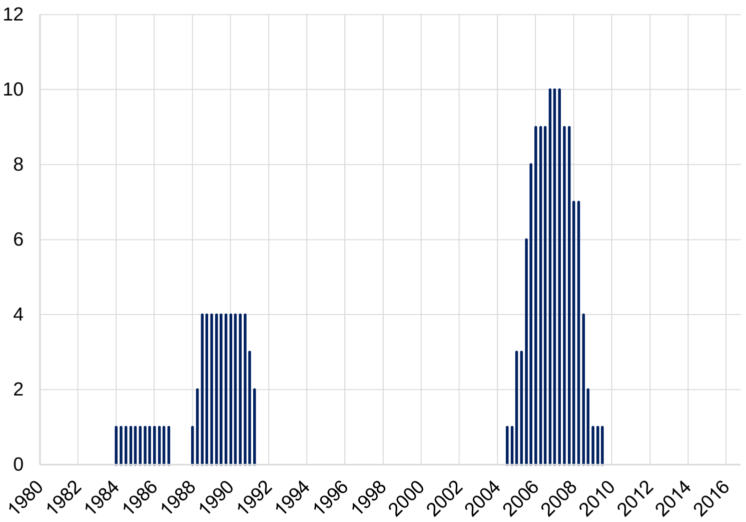
$$\Pr\left(y_{i,t} = 1 \middle| \alpha, X_{K,i,t}\right) = F\left(\alpha + X'_{K,i,t}\beta_K\right)$$

- $y_{i,t}$ denotes our response variable (taking the value 1 for observations 12 to 5 quarters before the real estate crises and 0 otherwise)
- *i* stands for country
- $X'_{K,i,t}$ explanatory variables including a constant term
- $F\,(\bullet)$ represents a logistic function of the form $F\,(z) = (1 + e^{-z})^{-1}$
- We use robust standard errors

LHS crisis variable

- RRE-related crisis as identified in Ferrari, Cornacchia, Pirovano (2015) complemented with information from ESRB crisis database → 15 crisis events in EU countries and UK

Vulnerable state pre-RRE-related systemic banking crises
(number of countries in a vulnerable state at a given time)



- The dependent variable of the logit model is the vulnerable state pre-RRE-related systemic banking crises, defined as 12-5 quarters before a crisis event
- Dependent variable is set to missing during a systemic crisis
- If we could start the sample in 1980s, we would have 15 observations, 1/3 of them pre-global financial crisis (for the moment we only the use 7 observations in the multivariate regression – further details later)

RHS variables (I/II)

- Following the literature and expert judgment, we consider variables on credit developments, debt burdens, house prices (dynamics and overvaluation), real estate and construction activity and we divide them into following categories:
  - Structural credit and debt burden (levels)
  - Cyclical credit and debt burdens (changes)
  - RRE prices
  - Construction/real estate investment
- Following Lang et al (2019), we consider the following transformations of variables:
  - 1 quarter, 1 year, 2-year and 3-year changes in percentage points
  - 1 quarter, 1 year, 2-year and 3-year growth rates
  - Gaps
- Putting together the raw variables and their transformations, we initially considered more than 100 variables in the univariate setting – after we exclude variable that are not available for any crisis in the 90s or not available for several countries we arrive at ca. 70 for the univariate analysis

RHS variables (II/II)

Variables	Transformations
Credit to NFPS as % GDP	1 quarter, 1 year, 2-year and 3-year changes in percentage points
Credit to NFPS	1 quarter, 1 year, 2-year and 3-year real growth rates
Credit to HHs as % of GDP	1 quarter, 1 year, 2-year and 3-year changes in percentage points
Credit to HHs	1 quarter, 1 year, 2-year and 3-year real growth rates
Credit to HHs	gap (400'000)
Credit to NFCs	gap (400'000)
Loans to HHs for house purchase	1 quarter, 1 year, 2-year and 3-year real growth rates
NFPS Debt service ratio	1 quarter, 1 year, 2-year and 3-year changes in percentage points
HHs Debt service ratio	1 quarter, 1 year, 2-year and 3-year changes in percentage points
House price to income	1 quarter, 1 year, 2-year and 3-year changes in percentage points
RRE prices	1 quarter, 1 year, 2-year and 3-year real growth rates
RRE prices	gap (400'000)
Gross value added contrustion activities	1 quarter, 1 year, 2-year and 3-year real growth rates
Gross value added of contruction and real estate activitie	1 quarter, 1 year, 2-year and 3-year real growth rates
Share of GVA by construction and real estate activities	1 quarter, 1 year, 2-year and 3-year changes in percentage points
RRE investment	1 quarter, 1 year, 2-year and 3-year growth rates
RRE investment to GDP	1 quarter, 1 year, 2-year and 3-year real growth rates

Univariate results in-sample

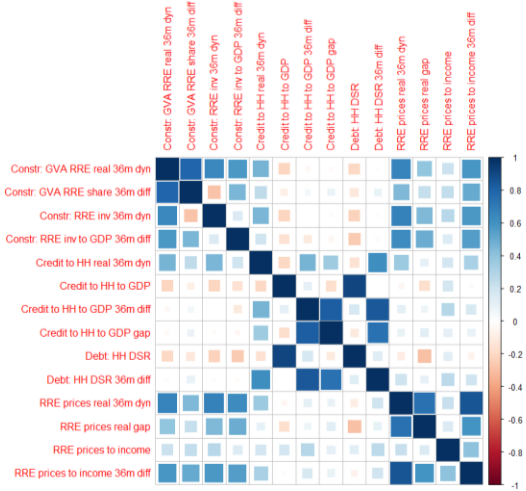
Variable	Coefficient	AUROC	Category
HH debt service ratio, 3-year change	1.627***	0.901	Cyclical credit and debt burden
Credit to HH as a % of GDP,3-year change	0.548***	0.855	Cyclical credit and debt burden
RRE prices to income, 3-year change	0.265***	0.850	RRE prices
Credit to HHs, 3-year real growth	0.0315**	0.822	Cyclical credit and debt burden
RRE prices, 3-year real growth	0.147***	0.809	RRE prices
Real RRE prices gap	0.0756***	0.787	RRE prices
Real estate and contruction GVA, 3 year growth	0.130***	0.770	Construction and real estate activity
RRE investment, 3-year growth	0.0265***	0.763	Construction and real estate activity
Credit to HHs as % GDP gap	0.222***	0.759	Credit
RRE investment as % GDP, 3-year growth	6.760***	0.746	Construction and real estate activity
Share of GVA from construction and real estate, 3-year growth	168.5***	0.717	Construction and real estate activity
RRE prices to income	0.0361**	0.697	RRE prices
HH debt service ratio	0.111**	0.661	Structural credit and debt burden
HH credit to GDP	0.0220**	0.658	Structural credit and debt burden

- In the univariate analysis we use all crises available for the different variables → we do not restrict the sample of variables to a common sample of crisis (as opposed to the multivariate)
- We see that:
  - Credit developments and debt burden variables for the HH sector perform (slightly) better than credit to NFPS
  - Transformations over longer horizons (36 months) perform better than over shorter horizons – persistence in the trends is relevant
- Univariate analysis is used to select the best performing variables for the multivariate analysis - we select several top variables from each 4 categories (on the basis of in-sample AUROC and statistical significance of coefficients)
- We do not consider similar transformations of the same variable (e.g. 1 credit to GDP change)

Multivariate (I/II) – initial choices

- Limited data availability implies that we face a trade-off between more LHS and more RHS variables
- Multivariate analysis based on 7 crisis (including one in the 90s) and 14 RHS variables
- We have to consider high pair-wise correlations between variables for the multivariate analysis
- We estimate multivariate models on different combinations of 4 type of variables (including 1 variable per category)

Correlations between variables selected in the univariate exercise



Selection of variables

Structural	Cyclical	RRE prices	Construction and real estate
Credit to HH as % GDP	HH debt service ratio, 3-year change	RRE prices to income, 3-year change	Real estate and contruction GVA. 3 year growth
HH debt service ratio	Credit to HH as a % of GDP,3-year change	RRE prices, 3-year real growth	RRE investment, 3-year growth
	Credit to HHs, 3-year real growth	Real RRE prices gap	RRE investment as % GDP, 3-year change
	Credit to HHs as % GDP, gap	RRE prices to income	Share of GVA from construction and real estate, 3-year change

Multivariate model in-sample (II/II) – top ten results

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	crisrre_ pc12t5	crisrre_p c12t5	crisrre_p c12t5	crisrre_ pc12t5	crisrre_p c12t5	crisrre_ pc12t5	crisrre_p c12t5	crisrre _pc12t	crisrre_p c12t5	crisrre_ pc12t5
	In-sample	In-sample	In-sample	In-sample	In-sample	n-sample	In-sample	n-sampl	In-sample	n-sample
	logit73	logit76	logit69	logit9	logit74	logit5	logit72	logit75	logit65	logit70
VARIABLES	Counter 1	Counter 2	Counter 3	Counter 4	Counter 5	Counter 6	Counter 7	Counter 8	Counter 9	Counter 10
Structural credit and debt burdens										
dsrhh	0.225**	0.196**	0.241**		0.178*		0.216**	0.172*	0.222**	0.194**
cthh_2gdp				0.0402*		0.0452**				
Cyclical credit and debt burdens										
dsrhh_d36m	0.845**	0.924***	0.900***	1.118***	0.975**	1.163***	0.880***	0.995**	0.802***	0.936***
RRE prices										
rre_hpi_d36m									0.181**	
rrep_r_gapr400	0.0625*	0.0718**		0.0577	0.0656*			0.0664		
rrep_r_g36m			0.110			0.0954	0.162***			0.130*
Construction and real estate										
gva_constr_re_r_g36m	0.152**		0.184**	0.125*		0.169*			0.119	
share_gva_constr_re_d36m		107.8					101.5			
rre_inv_g36m					0.0216					0.0373
rre_inv_2gdp_d36m								1.429		
Constant	-8.144***	-7.391***	-8.573***	-7.456***	-7.152***	-7.961***	-7.850***	-6.917***	-7.887***	-7.555***
Observations	1,242	1,242	1,230	1,242	1,237	1,230	1,230	1,237	1,232	1,225
Pseudo R2	0.471	0.458	0.463	0.464	0.448	0.461	0.446	0.447	0.448	0.438
AUROC	0.941	0.936	0.935	0.932	0.931	0.930	0.930	0.930	0.928	0.928

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

- Debt service ratio and the 3-year changes in the debt service ratio to HHs are the variables that exhibit the best performance in the multivariate setting
- However, credit, RRE prices and construction and real estate activity variables not always significant or not selected in the top specification
- To-do - one robustness check would be to re-run the multivariate analysis using variables selected on the basis of a univariate analysis using the same sample (data series length) as used for the multivariate analysis
- To-do – Aggregate the model results in an index

Key takeaways and way forward

- EWM exercise aims at detecting vulnerable state pre-RRE-related systemic banking crises
  - High AUROC in-sample
  - Top variables relate to credit developments, debt burden and house prices
  - Transformations over longer horizons (36 months) perform better than over shorter horizons
- Next steps and key challenges:
  - Limited data availability in our macroprudential database imply that we lose many observations (i.e. crises) – should we try to improve the data availability by using other sources (OECD or national)?
  - Current exercise is only in-sample – out-of-sample will be very challenging due to number of crises, but we could exclude one crises at a time and test out-of-sample properties when trying to detect this vulnerable state
  - Aggregation of best performing variables in an index using weights from the multivariate models and show its early warning properties
  - Possible overlap / correlation with the broader SRI as a key challenge to be addressed

ANNEX

LHS variable - RRE Crises considered

Country	Start date 1	End date 1	Start date 2	End date 2
DK	1987q1	1995q1	2008q1	2013q4
FI	1991q3	1996q4		
FR	1991q2	1995q1		
HU	2008q3	2010q3		
IE	2008q3	2013q4		
LV	2008q4	2010q3		
LT	2008q4	2009q4		
NL	2008q1	2013q1		
SI	2009q4	2014q4		
ES	2009q1	2013q4		
SE	1991q1	1997q2	2008q3	2010q4
GB	1991q3	1994q2	2007q3	2010q1

Univariate in-sample results – all variables

Variable	coefficient	AUROC	category
gva_constr_re_r_g36m	0.130***	0.77	construction and real estate activity
rre_inv_g36m	0.0265***	0.76	construction and real estate activity
gva_constr_r_g36m	0.0779***	0.75	construction and real estate activity
rre_inv_2gdp_d36m	6.760***	0.75	construction and real estate activity
gva_constr_re_r_g24m	0.112***	0.73	construction and real estate activity
share_gva_constr_re_d36m	168.5***	0.72	construction and real estate activity
rre_inv_g24m	0.0262***	0.71	construction and real estate activity
gva_constr_r_g24m	0.0708***	0.70	construction and real estate activity
rre_inv_2gdp_d24m	3.716*	0.69	construction and real estate activity
share_gva_constr_re_d24m	112.1***	0.68	construction and real estate activity
rre_inv_2gdp	0.998***	0.67	construction and real estate activity
gva_constr_r	0.0318*	0.65	construction and real estate activity
rre_inv	0.0330	0.64	construction and real estate activity
gva_constr_re_r_g12m	0.0734***	0.63	construction and real estate activity
share_gva_constr_re_d12m	60.47***	0.62	construction and real estate activity
gva_constr_r_g12m	0.0382**	0.61	construction and real estate activity
rre_inv_g12m	0.0170**	0.60	construction and real estate activity
gva_constr_re_r	0.00775	0.60	construction and real estate activity
share_gva_constr_re	0.160	0.59	construction and real estate activity
rre_inv_2gdp_d12m	0.494	0.56	construction and real estate activity
share_gva_constr_re_d3m	7.183	0.52	construction and real estate activity
gva_constr_re_r_g3m	0.00870	0.51	construction and real estate activity
rre_inv_g3m	0.000893	0.51	construction and real estate activity
gva_constr_r_g3m	0.000796	0.50	construction and real estate activity
rre_inv_2gdp_d3m	-0.166	0.50	construction and real estate activity

Univariate in-sample results – all variables

Variable	coefficient	AUROC	category
cthh_2gdp_d36m	0.548***	0.86	Credit
cbnfps_2gdp_d24m	0.255***	0.85	Credit
cbnfps_2gdp_d36m	0.279***	0.85	Credit
cbnfps_r_g36m	0.0684***	0.83	Credit
cbnfps_r_g24m	0.0780***	0.83	Credit
cthh_2gdp_d24m	0.443***	0.82	Credit
cthh_r_g36m	0.0315**	0.82	Credit
cbnfps_2gdp_d12m	0.205***	0.82	Credit
cbnfps_2gdp_gap400	0.109***	0.80	Credit
loanhp_hh_ea_s_r_g36m	0.0534*	0.79	Credit
cthh_r_g24m	0.0364***	0.79	Credit
loanhp_hh_ea_s_r_g24m	0.0395	0.78	Credit
cbnfps_r_g12m	0.0738***	0.77	Credit
cthh_2gdp_gap400	0.222***	0.76	Credit
cthh_2gdp_d12m	0.304***	0.76	Credit
cbnfps_2gdp_d3m	0.438***	0.76	Credit
cbnfps_2gdp	0.0153**	0.73	Credit
loanhp_hh_ea_s_r_g12m	0.0345	0.72	Credit
cthh_r_g12m	0.0356***	0.71	Credit
cbnfps_r_g3m	0.183***	0.69	Credit
cthh_2gdp_d3m	0.664***	0.68	Credit
cthh_2gdp	0.0220**	0.66	Credit
loanhp_hh_ea_s_r_g3m	0.0528	0.66	Credit
cbnfps_r	0.000558**	0.64	Credit
cthh_r_g3m	0.107***	0.63	Credit
cthh_r	0.000681**	0.62	Credit
loanhp_hh_ea_s_r	0.000262	0.47	Credit

Univariate in-sample results – all variables

Variable	coefficient	AUROC	category
dsrhh_d36m	1.627***	0.90	Debt burdens
dsrhh_d24m	1.862***	0.88	Debt burdens
dsrnfps_d24m	0.734***	0.85	Debt burdens
dsrnfps_d12m	0.967***	0.83	Debt burdens
dsrhh_d12m	2.397***	0.82	Debt burdens
dsrnfps_d36m	0.540***	0.81	Debt burdens
dsrnfps_d3m	1.393***	0.74	Debt burdens
dsrhh_d3m	4.402***	0.71	Debt burdens
dsrhh	0.111**	0.66	Debt burdens
dsrnfps	0.0311	0.65	Debt burdens

Univariate in-sample results – all variables

Variable	coefficient	AUROC	category
rre_hpi_d36m	0.265***	0.85	real estate prices
rrep_r_g36m	0.147***	0.81	real estate prices
rre_hpi_d24m	0.172***	0.80	real estate prices
rrep_r_gapr400	0.0756***	0.787	real estate prices
rrep_r_g24m	0.100***	0.75	real estate prices
rre_hpi	0.0361**	0.70	real estate prices
rrep_r	0.0241**	0.69	real estate prices
rre_hpi_d12m	0.0797***	0.68	real estate prices
rrep_r_g12m	0.0483***	0.64	real estate prices
rre_hpi_d3m	0.0876***	0.56	real estate prices
rrep_r_g3m	0.0517	0.55	real estate prices

Aggregation of results

- The RRE-index could be a weighted average of the individual “normalised” top performing indicators

$$RRE\ index_{i,t} = \sum_{j=1}^{K} \omega^j \cdot \tilde{x}_{i,t}^j$$

$$\tilde{x}_{i,t}^j = \left(x_{i,t}^j - x_M^j\right) / x_{IQR}^j$$

Weights

- Common weights across countries and time
- Weights chosen to optimise early warning properties of the SRI

Normalisation

- Common normalisation for all countries based on pooled median and IQR
- Normalisation provides for better interpretation of RRE index units and weights
- <u>Normalisation does not affect the index properties </u>compared to using raw data